(iii) $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \hat{i} - 4\hat{j} + \hat{k}$

Direction cosines are $\frac{1}{\sqrt{18}}, \frac{-4}{\sqrt{18}}, \frac{1}{\sqrt{18}}$.

(iv) Let A and B be the points (2, 3, 1) and (3,–1, 2). The direction cosines of $\overrightarrow{AB}$ are $\frac{1}{\sqrt{18}}, \frac{-4}{\sqrt{18}}, \frac{1}{\sqrt{18}}$.

But any point can be taken as first point. Hence we have another set of direction cosines with opposite direction. Thus, we have another set of direction ratios $\frac{-1}{\sqrt{18}}, \frac{4}{\sqrt{18}}, \frac{-1}{\sqrt{18}}$.

(v) The direction cosines are $\frac{2}{7}, \frac{3}{7}, \frac{6}{7}$.

The unit vector is $\frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k}$.

The required vector is $\frac{5}{7}(2\hat{i} + 3\hat{j} + 6\hat{k})$.

Example 8.7

Show that the points whose position vectors are $2\hat{i} + 3\hat{j} - 5\hat{k}$, $3\hat{i} + \hat{j} - 2\hat{k}$ and , $6\hat{i} - 5\hat{j} + 7\hat{k}$ are collinear.

Solution

Let O be the origin and let $\overrightarrow{OA}, \overrightarrow{OB}$, and $\overrightarrow{OC}$ be the vectors $2\hat{i} + 3\hat{j} - 5\hat{k}$, $3\hat{i} + \hat{j} - 2\hat{k}$ and $6\hat{i} - 5\hat{j} + 7\hat{k}$ respectively. Then

$$\overrightarrow{AB} = \hat{i} - 2\hat{j} + 3\hat{k} \text{ and } \overrightarrow{AC} = 4\hat{i} - 8\hat{j} + 12\hat{k}.$$

Thus $\overrightarrow{AC} = 4\overrightarrow{AB}$ and hence $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are parallel. They have a common point namely A. Thus, the three points are collinear.

Alternative method

Let O be the point of reference.

Let $\overrightarrow{OA} = 2\hat{i} + 3\hat{j} - 5\hat{k}, \overrightarrow{OB} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\overrightarrow{OC} = 6\hat{i} - 5\hat{j} + 7\hat{k}$

$$\overrightarrow{AB} = \hat{i} - 2\hat{j} + 3\hat{k} \ ; \ \overrightarrow{BC} = 3\hat{i} - 6\hat{j} + 9\hat{k} \ ; \ \overrightarrow{CA} = -4\hat{i} + 8\hat{j} - 12\hat{k}$$

$$\left|\overrightarrow{AB}\right| = \sqrt{14} \ ; \ \left|\overrightarrow{BC}\right| = \sqrt{126} = 3\sqrt{14} \ ; \ \left|\overrightarrow{CA}\right| = \sqrt{224} = 4\sqrt{14}.$$

Thus, $AC = AB + BC$.

Hence A, B, C are lying on the same line. That is, they are collinear.

Example 8.8

Find a point whose position vector has magnitude 5 and parallel to the vector $4\hat{i} - 3\hat{j} + 10\hat{k}$.

Solution :

Let $\vec{a}$ be the vector $4\hat{i} - 3\hat{j} + 10\hat{k}$.

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